Surface response to stratospheric aerosol changes in a coupled atmosphere–ocean model

J. C. Fyfe,[1] K. von Salzen,[1] J. N. S. Cole,[1] N. P. Gillett,[1] and J.-P. Vernier[2,3]

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[1] Previous work with a simple climate model has suggested a global cooling impact of increasing stratospheric aerosol. Here we use a comprehensive Earth System Model including coupled atmosphere and ocean components to show that increasing stratospheric aerosol since the late 1990s has reduced global warming by at least 0.07 C to present and that a further global cooling impact will occur if the observed stratospheric aerosol trend continues to the end of this decade. This result confirms the previous work and suggests that climate models that do not account for stratospheric aerosol increase will overestimate global warming to a small but notable degree. An additional new finding is that increasing stratospheric aerosol since the late 1990s has reduced the rise in global mean precipitation. [2] Finally, regional patterns of change in simulations with stratospheric aerosol increase to year 2020 show ~40% less equatorial precipitation increase and ~60% greater surface pressure decrease around Antarctica, relative to simulations without such stratospheric aerosol changes. **Citation:** Fyfe J. C., K. von Salzen, J. N. S. Cole, N. P. Gillett, and J.-P. Vernier (2013), Surface response to stratospheric aerosol changes in a coupled atmosphere–ocean model, *Geophys. Res. Lett.*, *40*, 584–588, doi:10.1002/grl.50156.

1. Introduction

[3] *Solomon et al.* [2011] present results from several independent datasets showing that stratospheric aerosol abundance has increased since 2000—due, it is thought, to a series of mild tropical eruptions, as opposed to increasing anthropogenic emissions (*Vernier et al.* [2011]). *Solomon et al.* [2011] show that this increase in stratospheric aerosol implies a negative radiative forcing over the period of about -0.1 W m^{-2}. Using the Bern 2.5cc intermediate complexity climate model (*Plattner et al.* [2008]), *Solomon et al.* [2011] also show that the observed increase in stratospheric aerosol caused global cooling of about 0.07 C over the period from 2000 to 2010 as compared with a model simulation in which no stratospheric aerosol increase was assumed after year 2000. Since global climate models, such as those involved in Phase 5 of the Coupled Model Intercomparison Project (CMIP5), generally neglect the radiative forcing associated with increasing stratospheric aerosol after year 2000, this implies that they have overestimated global warming over this period and will continue to do so if these stratospheric aerosols continue to increase. That said, *Solomon et al.* [2011] acknowledge that the stratospheric aerosol effect that they uncovered is small and "could be difficult to quantify in Atmosphere–ocean General Circulation Models (AOGCMs) against the computed noise of internal variability," which the Bern 2.5cc model cannot represent. Using an Earth System Model with coupled atmosphere and ocean components, we confirm the findings of *Solomon et al.* [2011] and provide new evidence of global cooling and drying impacts from increasing stratospheric aerosol since the late 1990s.

2. Model and Simulations

[4] The second-generation Canadian Earth System Model (CanESM2) is a comprehensive ESM which includes interactions between the terrestrial and oceanic carbon cycle components and the physical climate system (*Arora et al.* [2011]). The atmospheric component of CanESM2 is a spectral model employing T63 triangular truncation with physical tendencies calculated on a 128×64 ($\sim 2.81°$) horizontal linear grid and based on the Canadian Centre for Climate Modelling and Analysis (CCCma) fourth-generation atmospheric general circulation model (CanAM4; *von Salzen et al.* [2012]). CanAM4 has 35 unevenly spaced vertical levels up to ~0.1 hPa. The physical ocean component of CanESM2 is based on the National Center for Atmospheric Research (NCAR) community ocean model (NCOM1.3) and has 40 levels with approximately 10 m resolution in the upper ocean, and the horizontal resolution is approximately 1.41° (longitude) × 0.94° (latitude).

[5] Baseline simulations up to 2005 were forced with changes in greenhouse gases, aerosols (sulphate, black carbon, and organic carbon), ozone (tropospheric and stratospheric), land use change (e.g., deforestation), and volcanic and solar variability.

[6] The stratospheric optical depth data used in our baseline simulations was, as in most other CMIP5 simulations (*Driscoll et al.* [2012]), that of *Sato et al.* [1993] which was prescribed in four latitude bands and extended using exponential decay to near zero after 2000 (see the black curve in Figure 1a). Therefore, as with the other simulations contributed to CMIP5, there is effectively no stratospheric aerosol forcing after 2000 in our baseline simulations using the standard CMIP forcings. Here we note that the baseline stratospheric optical depth used in *Solomon et al.* [2011] was taken from a more recent version of the *Sato et al.* [1993] dataset which shows a somewhat slower decay to zero near 2000 (see the blue curve in Figure 1b) than in

All Supporting Information may be found in the online version of this article.

[1]Canadian Centre for Climate Modelling and Analysis, Environment Canada, Victoria, British Columbia, Canada.
[2]Science Systems and Applications, Inc, Hampton, Virginia, USA.
[3]NASA Langley Research Center, Hampton, Virginia, USA.

Corresponding author: J. C. Fyfe, Canadian Centre for Climate Modelling and Analysis, University of Victoria, P.O.Box 1700 STN CSC, Victoria, British Columbia, Canada, V8W 2Y2. (john.fyfe@ec.gc.ca)

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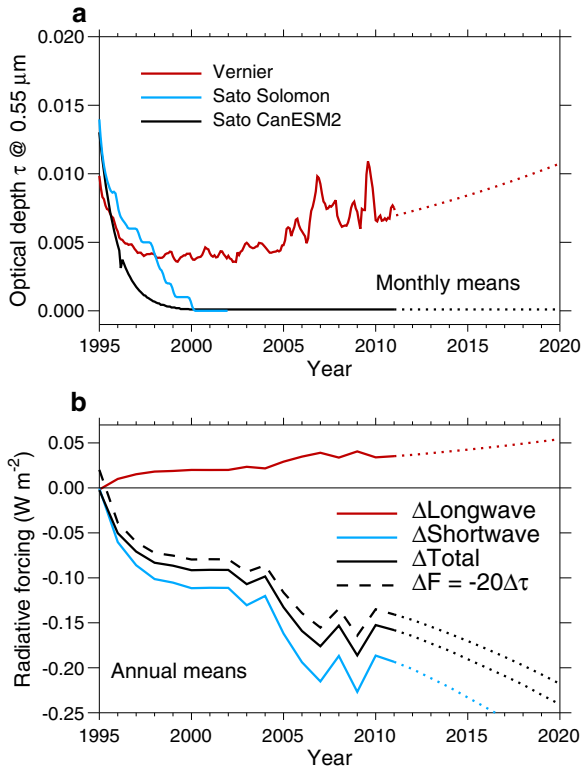

Figure 1. Stratospheric aerosol optical depth (a) and radiative forcing difference between the datasets of *Vernier et al.* [2011] and *Sato et al.* [1993] (b). Radiative forcings were estimated using two approaches. In the first approach, the instantaneous radiative forcing (solid curves) at a climatological tropopause was obtained by calling the radiative transfer code twice in CanESM2 and then finding the difference in the results. The first call used the dataset of *Vernier et al.* [2011], and the second call used the dataset of *Sato et al.* [1993]. This forcing difference drove the climate response discussed in our paper. In the second approach, the radiative forcing (dashed curve) was computed using the relationship $F = -20\tau$ between radiative forcing F and stratospheric aerosol optical thickness τ. This approach is similar to that in the study by *Solomon et al.* [2011].

our baseline simulations. Our baseline simulations were merged with simulations using the standard RCP4.5 forcing scenario from the period 2006 to 2020, noting that the RCP4.5 simulations also had close to zero stratospheric aerosol. An analysis of the surface climate evolution in these simulations can be found in *Arora et al.* [2011].

[7] An ensemble of five baseline simulations (with differing initial conditions on 1 January 1850) was available and was compared to an ensemble of five simulations (initiated from a differing baseline simulations on 1 September 1995) with aerosol optical depth from the study by *Vernier et al.* [2011] prescribed (horizontally and vertically uniform between the climatological tropopause and 10 hPa) from 1995 to 2011 and then increased by 5% per year from 2012 to 2020 (see the red curve in Figure 1a). These simulations were initiated at the month of crossover between the datasets of *Vernier et al.* [2011] and *Sato et al.* [1993] (red and black curves in Figure 1a).

[8] In contrast, *Solomon et al.* [2011] initiated their experiments in 1998 at the point of crossover between the datasets of *Vernier et al.* [2011] and *Sato et al.* [1993] (red and blue curves in Figure 1a). An additional ensemble of five CanESM2 simulations initiated in 1998 (not shown) indicate that our results are reasonably insensitive to the initiation date.

[9] The statistical significance of the ensemble mean differences obtained in our study are assessed using a two-sample *t* test (*von Storch and Zwiers* [1999]) based on a sample size of five in both samples, and with pooled standard deviation estimated from a 300 year preindustrial simulation with no external or natural (e.g., volcanic or solar) forcing changes.

3. Results

3.1. Radiative Forcing

[10] We begin by considering estimates of the radiative forcing associated with stratospheric aerosol changes. *Solomon et al.* [2011] estimated such radiative forcing using stratospheric aerosol optical thickness data from *Vernier et al.* [2011] in the relationship $F = -25\tau$ between adjusted radiative forcing F and optical thickness τ. The "-25" multiplier in this relationship is based on calculations using the NASA Goddard Institute for Space Studies (GISS) global climate model (GCM) in which a uniform stratospheric aerosol layer with optical thickness of 0.1 and aerosol effective radii of 0.51 μm was prescribed (*Hansen et al.* [2005]). In this way, *Solomon et al.* [2011] estimated a negative radiative forcing due to stratospheric aerosol changes from 2000 through 2010 of about -0.1 Wm^{-2} (see their Figure 3).

[11] Here we provide two additional radiative forcing estimates using CanESM2. Our first estimate uses the fixed sea surface temperature (SST) method of *Hansen et al.* [2005] with a uniform stratospheric aerosol layer of optical thickness of 0.1 to infer the relationship between optical thickness and radiative forcing. Unlike the relationship used in *Solomon et al.* [2011] ours is derived using 1) CanESM2 instead of the GISS GCM and 2) an aerosol effective radius of 0.35 μm rather than 0.51 μm. An effective radius of 0.35 μm is approximately the average effective radius taken over all volcanic events in the GISS dataset (http://data.giss.nasa.gov/modelforce/strataer/) while an effective radius of 0.51 μm is more representative of the effective radius shortly after the 1992 eruption of Mount Pinatubo. Using an effective radius of 0.35 μm gives the relationship $F = -20\tau$, and hence an estimate of the difference in radiative forcing between the *Vernier et al.* [2011] and *Sato et al.* [1993] datasets (see the dashed curve in Figure 1b).

[12] Our second estimate is the instantaneous longwave, shortwave, and total radiative forcing at a climatological tropopause in CanESM2 due to the presence of the stratospheric aerosols, shown in Figure 1 as solid curves. Here we note a radiative effect associated with the absorption, and emission, of outgoing longwave radiation by the aerosols in the lower stratosphere, similar to explosive volcanic aerosol (*Robock* [2000]). Both our calculations yield a negative total radiative forcing due to stratospheric aerosol changes from 2000 through 2010 (of ~-0.095 (instantaneous) and ~-0.085 Wm^{-2} (fixed-SST)) that is slightly less than the value of ~-0.10 Wm^{-2} by *Solomon et al.* [2011]. The slight difference with that of *Solomon et al.* [2011] may be due to several factors including the fact that we use a different radiative transfer model, a different GCM (CanESM2 versus the GISS GCM), and slightly different assumptions about the

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aerosol size distribution as discussed above. These differences aside, here we ask if the impact of the changes in radiative forcing found in our model on global mean surface temperature can be seen through internal variability noise? Note that internal variability noise is not present, by construction, in the radiative forcing presented in Figure 1b or Figure 3 of *Solomon et al*. [2011] or *Solomon et al*. [2011] time series of global mean surface temperature (their Figure 4b).

3.2. Response in Surface Temperature

[13] Figure 2 compares the annual mean and global mean surface temperature anomalies in the simulations with stratospheric aerosol changes (thick pink curve) and without stratospheric aerosol changes (thick black curve). In both cases, these time series represent averages over five ensemble members—a mathematical operation that reduces internal variability noise and should make the impact of stratospheric aerosol change more apparent. Obvious in these time series is the cooling impact of explosive volcanic eruptions, which we separate into the dashed curve in Figure 2 using the method of *Thompson et al*. [2009] (see also *Fyfe et al*. [2010]). In Figure 2, we see some degree of separation between the thick black curve (without stratospheric aerosol changes) and the thick pink curve (with stratospheric aerosol changes), although the separation is small (e.g., much smaller that the cooling effect of explosive volcanic eruptions) and does not stand out above ± one standard deviation of internal interannual variability shown with gray shading. In our model, and others (*Fyfe et al*. [2010]), internal interannual variability in global mean surface temperature is dominated by signals of dynamically induced atmospheric variability and the El Niño-Southern Oscillation (ENSO)—quantified here by the fact that about 75% of the standard deviation shown in the gray shading in Figure 2 can be attributed to these two signals (as estimated using the approach by *Thompson et al*. [2009]).

[14] In short, the impact of stratospheric aerosol change is small and difficult to discern against interannual variability noise, although not impossible as we will see next.

[15] A clearer signal of stratospheric aerosol change is seen in decadal average time series of global mean surface temperature, as shown in the thin black (without stratospheric aerosol change) and thin pink (with stratospheric aerosol change) curves in Figure 2.

[16] Based on the difference between these decadal average time series we conclude that the observed increase in stratospheric aerosol since the late 1990s caused a global cooling impact of -0.07±0.07 C over the 2002–2012 period (using 95% confidence intervals), relative to the simulations without stratospheric aerosol change (Figure 3).

[17] This estimate is very close to that obtained by *Solomon et al*. [2011] using a simple climate model with no internal atmospheric variability.

[18] Assuming stratospheric aerosols continue to increase following the recent historical trend, our model projects a continuation of this global cooling impact. Our key result then is that with sufficient averaging (ensemble and time), we find a significant signal of stratospheric aerosol change in global mean surface temperature. (Here we note that the cooling impact is felt most significantly in the region between about 30 N and 30 S, as shown in Figure 4a.) We stress though that the global cooling impact seen here is

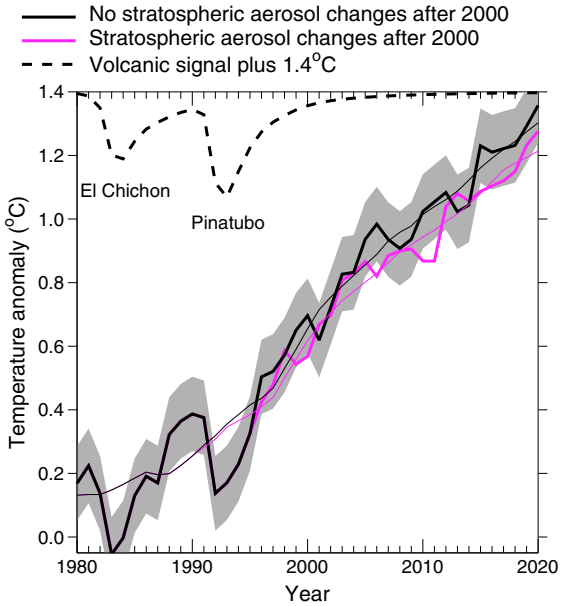

Figure 2. Global mean surface temperature anomalies. Anomalies are relative to 1961–1990 averages. The thick black (thick pink) curve is the ensemble and annual average of simulations without (with) changes in stratospheric aerosol. Gray shading about the thick black curve is ± one standard deviation of the annual mean anomalies derived from a 300 yearlong preindustrial simulation with no external or natural forcing changes. Dashed curve is the signal associated with explosive volcanic eruptions in the simulation with no stratospheric aerosol changes computed following *Thompson et al.* [2009] (note the offset of 1.4 C). The thin black (thin pink) curve is the ensemble and running decadal average of simulations without (with) changes in stratospheric aerosol.

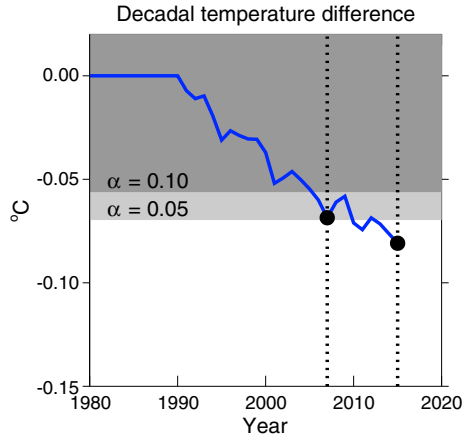

Figure 3. Decadal mean and global mean surface temperature difference between simulations with changes in stratospheric aerosol and simulations without. "Year" represents the central year in an 11-year running average. For example, the difference at year 2007, indicated by the leftmost black dot, is the average difference over the historical period from 2002 to 2012. Similarly, the difference at year 2015, indicated by the rightmost black dot, is the average difference over the (mostly) future period from 2010 to 2020. Light and dark gray shadings are 95% and 90% ranges of preindustrial control variability.

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small compared to the ~0.8 C warming simulated from the late 1990s to year 2020 with greenhouse gas and other anthropogenic forcing changes following the RCP4.5 scenario. That said, the relative impact of stratospheric aerosol increase on some other climatic variables is more substantial, as we shall see next.

3.3. Response in Other Surface Variables

[19] A number of studies show a decrease in global precipitation following volcanic eruptions (e.g., *Robock* [2000]; *Gillett et al*. [2004] and references contained). Decreased global precipitation has also been shown in geoengineering experiments where CO_2-induced global warming is offset by reducing incident solar radiation (e.g., *Govindasamy et al*. [2003]) or by injecting sulphate particles into the stratosphere (e.g. *Fyfe et al*. [2013]).

[20] Our simulations similarly show that changes in stratospheric aerosol impact global mean precipitation. Specifically, simulations with stratospheric aerosol increase since the late 1990s show slightly less global mean precipitation increase (significant at just less than the 90% level) than simulations without such stratospheric aerosol increase. A global drying impact of about -0.008±0.005 mm day^{-1} (using 95% confidence levels) is seen in simulations to year 2020 with continuing stratospheric aerosol increase. To account for such precipitation change, we consider the impact of increasing stratospheric aerosol on the global energy budget (see Supporting Information). At the top of the atmosphere, decadal average differences show that increased outgoing shortwave radiation due to the albedo increase is effectively balanced by decreased upwelling longwave radiation due to cooling of the surface and troposphere and absorption by the stratospheric aerosol. At the surface, the stratospheric aerosol increase causes a reduction in net radiative flux, due to comparable longwave and shortwave cooling contributions, which is nearly in balance with a reduction in latent heat flux, and hence the global mean precipitation reduction.

[21] The pattern of zonal mean precipitation change due to stratospheric aerosol increase from the late 1990s to the end of this decade is shown in Figure 4b (blue curve). This pattern indicates reduced tropical moistening and reduced subtropical drying compared to the change in simulations without such stratospheric aerosol increase (Figure 4b, black curve). In other words, the stratospheric aerosol impact opposes the well-known pattern of surface tropical moistening and subtropical drying that is simulated in response to increasing anthropogenic greenhouse gas concentrations in the atmosphere (*IPCC* [2007]). In terms of magnitude, the increase in stratospheric aerosol causes equatorial moistening to be reduced by more than 40% in our simulations that encompass observed and future projected stratospheric aerosol increase. Here we note that the pattern of stratospheric aerosol-induced precipitation change shown in Figure 4b is not clear when the computation is for the historical period up to

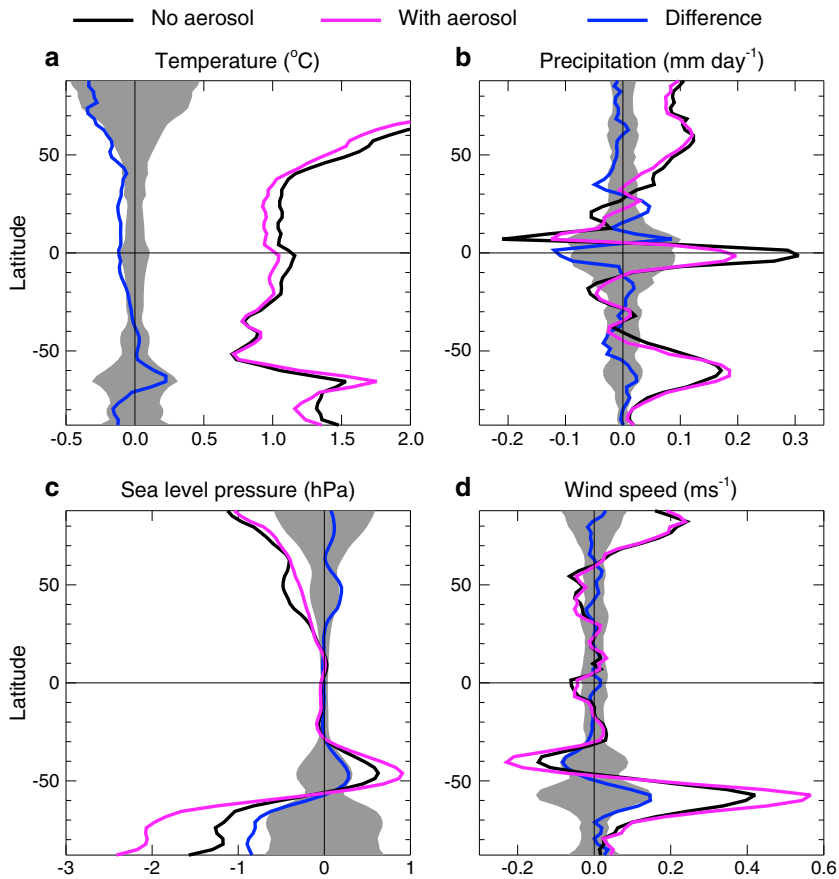

Figure 4. Decadal mean and zonal mean difference between 1950–1960 and 2010–2020 for simulations without stratospheric aerosol increase (black curve) and with such increase (pink curve). The blue curve is the difference between the simulations. a. Surface temperature. b. Precipitation. c. Sea level pressure. d. Surface wind speed. Gray shading is the 90% confidence level.

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2012 and hence depends on our assumption of continued stratospheric aerosol increase.

[22] Another impact of stratospheric aerosol increase is on Southern Hemisphere extratropical surface circulation. As seen in Figure 4c (pink curve), increasing stratospheric aerosol causes higher sea level pressure (SLP) in southern mid-latitudes (between about 30 S and 60 S) and lower SLP in high latitudes (between about 60 S and 90 S), as compared to simulations without stratospheric aerosol changes (black curve). The pattern of SLP response (blue curve) is reminiscent of the high polarity pattern of internal variability associated with the Southern Annular Mode (SAM) (*Thompson and Wallace* [2000]) and is consistent with the positive SAM pattern simulated in response to volcanic eruptions (*Karpechko et al*. [2010]) and stratospheric aerosol injection in geoengineering experiments (*McCusker et al*. [2012]). Notable is the fact that the decrease in SLP around Antarctic, at ~65 S, is about 60% larger in the simulations that include the stratospheric aerosol effect than in those that do not.

[23] The stratospheric aerosol-induced SLP pattern shown in Figure 4c reflects strengthened westerlies around Antarctica (Figure 4d) which may affect many aspects of the climate there, including ocean circulation (e.g., *Fyfe and Saenko* [2006]), ocean carbon uptake (e.g., *Le Quéré et al*. [2007] and *Zickfeld et al*. [2007]), and Antarctic sea ice (e.g., *Sigmond and Fyfe* [2010]), although these impacts are not explored here.

4. Discussion

[24] In an earlier study using a simple climate model, it was shown that increasing stratospheric aerosol since the late 1990s had a cooling impact on global mean surface temperature (*Solomon et al*. [2011]). Here we have confirmed this result, in sign and magnitude, using an Earth System Model with coupled atmosphere and ocean components. Specifically, our simulations show that increasing stratospheric aerosol has reduced global warming by more than ~-0.07 C to present and may reduce it further if the observed stratospheric aerosol trend continues. Here it is worth noting that the CMIP5 simulations generally did not account for any stratospheric aerosol increase since the late 1990s; hence the CMIP5 models may have overestimated global temperature change, but that overestimation is probably only about −0.07 C over the 2002–2012 period.

[25] We have also shown that increasing stratospheric aerosol has caused, and may continue to cause, reduced amounts of precipitation particularly in the tropics going forward, relative to the situation without increasing stratospheric aerosol. Assuming that stratospheric aerosols continue to increase, our model projects significantly lower surface pressures around Antarctica than obtained in simulations without a stratospheric aerosol increase, and this implies (through enhanced surface winds) an impact on ocean temperatures, ocean carbon uptake, and Antarctic sea ice. In short, climate models that do not account for stratospheric aerosol increase will overestimate global surface warming and moistening and underestimate southern extratropical circulation change.

[26] It is worth mentioning that the stratospheric aerosol-induced global mean surface temperature response that we find, while of a magnitude consistent with that of *Solomon et al*. [2011], is not statistically significant in any of the individual simulations that make up the ensemble average that we have considered, even to year 2020 under the assumption of continued increase in stratospheric aerosol. Therefore, the stratospheric aerosol effect considered here, and in the study by *Solomon et al*. [2011], is unlikely to be detectable in observations.

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